Water Quality Test Reports: Summary & Analysis

Vishnu Shivam Tower, Thakur Village, Kandivali East, Mumbai | Updated 31-10-2026 (adds the October 2026 kitchen water report) | Reference standard: IS 10500:2012 (Drinking Water)

Kitchen water, latest (Oct 2026): PASSES IS 10500 and is fit to drink. Coliform and E. coli are absent, and every chemical parameter is well within limits. The sample was collected by lab staff, so this result is more reliable than September's.

Borewell water (Dec 2025): still NOT fit to drink. Coliform and E. coli are present, turbidity is 216 NTU, and BOD and COD are high (organic or sewage pollution). It has not been retested since.
Kitchen water, Sep 2026: failed (Coliform and E. coli present). The October chemistry is very different from September's, which suggests borewell water was reaching the kitchen line in September (see section 3.4).

Dissolved Oxygen (DO) has not been tested in any of the three reports. Residual chlorine has not been tested either. Add both to the next test (see section 4).

1. Reports reviewed

Item	A: Borewell Water	B: Kitchen Water (Sep)	C: Kitchen Water (Oct): latest
Lab ID	VAL/2512/160070	VAL/2609/230075	VAL/2610/270216
ULR No.	TC1193925000035715F	TC1193926000039075F	TC1193926000045733F
Laboratory	Varni Analytical LLP, Mira Road (NABL accredited)		
Collected	16-12-2025	23-09-2026	27-10-2026
Collected by	Lab staff (Sudesh)	Sent by courier (society)	Lab staff (Dinesh)
Report date	22-12-2025	26-09-2026	30-10-2026
Parameters	21	17	17
Lab's verdict	**Does not pass IS 10500**	**Not fit for human consumption**	**Passes IS 10500; fit for human consumption**

2. Results side by side

Limits are given as acceptable / permissible. The acceptable limit is the one printed on the reports. The permissible limit is the relaxation IS 10500 allows when no alternative source exists. Result colours: **green = within acceptable**; **amber = over acceptable but within permissible**; **red = over permissible or not allowed at all**. ND = not detected.

Parameter	Unit	Limit (acc. / perm.)	A: Borewell Dec-25	B: Kitchen Sep-26	C: Kitchen Oct-26
Biological					
Total Coliform	/100 ml	Absent	**Present**	**Present**	**Absent**
E. coli	/100 ml	Absent	**Present**	**Present**	**Absent**
Physical					
Odour	–	Agreeable	**Disagreeable**	**Agreeable**	**Agreeable**
Taste	–	Agreeable	**Disagreeable**	**Agreeable**	**Agreeable**
Colour	Hazen	5 / 15	**10.00**	**ND<1.0**	**ND<1.0**
Turbidity	NTU	1 / 5	**216.00**	**1.00 (at limit)**	**ND<0.5**
pH	–	6.5–8.5	**7.69**	**7.49**	**7.34**
Conductivity	µS/cm	–	739.0	864.0	117.5
Chemical					
Total Dissolved Solids	mg/l	500 / 2000	**714.00**	**425.00**	**55.00**
Total Hardness (as $CaCO_3$)	mg/l	200 / 600	**190.00**	**210.00**	**37.00**
Total Alkalinity (as $CaCO_3$)	mg/l	200 / 600	**265.63**	**348.00**	**37.95**
Calcium	mg/l	75 / 200	**30.06**	**46.09**	**7.21**
Magnesium	mg/l	30 / 100	**27.95**	**23.09**	**4.62**
Chloride	mg/l	250 / 1000	**63.99**	**62.64**	**13.66**
Sulphate	mg/l	200 / 400	**15.74**	**17.36**	**ND<1.0**
Fluoride	mg/l	1.0 / 1.5	**ND<0.05**	**ND<0.05**	**ND<0.05**
Silica	mg/l	–	0.97	2.24	ND<0.01
Pollution indicators (not covered by IS 10500)					
Total Suspended Solids	mg/l	–	**178.00**	Not tested	Not tested
Chemical Oxygen Demand	mg/l	–	**24.79**	Not tested	Not tested
Biochemical Oxygen Demand	mg/l	–	**8.00**	Not tested	Not tested
Oil & Grease	mg/l	–	**ND<4.0**	Not tested	Not tested
Dissolved Oxygen	mg/l	–	Not tested	Not tested	Not tested

Score: **A (borewell)** has 6 failures and 3 results over acceptable. **B (kitchen, Sep)** has 2 failures (Coliform, E. coli) and 2 over acceptable (hardness, alkalinity). **C (kitchen, Oct)** has no failures and nothing over acceptable.

3. Analysis

3.1 Borewell water (December 2025)

- **Faecal contamination:** Coliform and E. coli are present, and BOD (8 mg/L) and COD (24.8 mg/L) are both raised. Clean groundwater normally has BOD below 2–3 mg/L. Together these point to **sewage or surface run-off reaching the borewell**, for example through a damaged casing, a missing sanitary seal, or a nearby drain or septic line.
- **Very high turbidity (216 NTU) and TSS (178 mg/L):** the water carries a lot of silt or suspended matter. That also explains the colour of 10 Hazen and the disagreeable odour and taste. Particles this fine shield bacteria from chlorine, so **the water must be filtered before it is disinfected**.
- **Minerals:** TDS of 714 mg/L is above the acceptable limit but well within the permissible 2000. Hardness and calcium are within limits. Magnesium (27.95) is close to its limit of 30. Fluoride, chloride and sulphate are low.
- **Data check:** the TDS-to-conductivity ratio is 714/739 = 0.97. The usual range is 0.55–0.75, so the TDS figure is probably inflated by fine colloidal matter. It is worth querying with the lab.
- **Suitability: not suitable for drinking or cooking.** After filtration and chlorination it may be used for flushing, gardening and washing. **Retesting is overdue**: the last test was more than 10 months ago.

3.2 Kitchen water, September 2026 (failed)

- Coliform and E. coli were present. Hardness (210 mg/L) and alkalinity (348 mg/L) were over the acceptable limits, and turbidity was right at the limit (1.00 NTU).
- The sample was **sent by courier** in a plastic bottle and not collected by the lab, so the bacteria result was less reliable. It still had to be treated as real.

3.3 Kitchen water, October 2026 (passed)

- **Bacteriology:** Total Coliform and E. coli are **absent**. The sample was collected by lab staff, so the result is reliable.
- **Chemistry:** every parameter is well inside the acceptable limits. Turbidity is below 0.5 NTU, TDS is 55 mg/L, hardness 37 mg/L and alkalinity 38 mg/L. This is **soft, low-mineral water**, typical of BMC's lake-fed municipal supply.
- **Data check:** calcium and magnesium give a calculated hardness of 18.0 + 19.0 = 37.0 mg/L as $CaCO_3$, which matches the reported 37.00 exactly. The TDS-to-conductivity ratio of 0.47 is normal for soft water. The anion and cation totals also balance reasonably, so the report is internally consistent.
- **Corrosion caution:** I estimate a Langelier Saturation Index of about −1.7 (at roughly 25 °C). Water like this tends to dissolve metal rather than deposit scale. It is not a health risk by itself, but it can slowly corrode old galvanised iron (GI) pipes and pick up iron or zinc, giving a rusty colour or metallic taste. If the building has old GI plumbing, test iron once (and lead if there are old fittings).

3.4 Kitchen trend, September to October: has the water source changed?

Parameter	Borewell (Dec-25)	Kitchen Sep-26	Kitchen Oct-26	Change Sep to Oct
Conductivity (µS/cm)	739	864	117.5	−86%
TDS (mg/l)	714	425	55	−87%
Total Hardness (mg/l)	190	210	37	−82%
Total Alkalinity (mg/l)	266	348	38	−89%
Calcium (mg/l)	30.1	46.1	7.2	−84%
Magnesium (mg/l)	28.0	23.1	4.6	−80%
Chloride (mg/l)	64.0	62.6	13.7	−78%
Sulphate (mg/l)	15.7	17.4	<1.0	>−94%

- Tank cleaning or disinfection does not change dissolved minerals by 80–90%. A drop this large means **the water itself was different**.
- The September kitchen sample looks like **groundwater**: high hardness and alkalinity, and chloride almost identical to the borewell (62.6 vs 64.0 mg/L). The October sample looks like **municipal (BMC) supply**.
- **Most likely explanation:** in September, borewell water was entering the kitchen or drinking-water line through a shared tank, a cross-connection or a valve left open. Since the borewell itself contained E. coli, this probably also caused September's bacterial failure.
- **Action:** find out what changed between 23-09 and 27-10 (tank cleaning, borewell pump switched off, valves changed, tank connections). Make that separation **permanent**, with physically separate tanks and lines, so borewell water can never reach drinking taps again.

4. Oxygen in water: Dissolved Oxygen (DO)

Status: not tested. DO does not appear in any of the three reports. The IS 10500 "partial" package used by the lab does not include it. It has to be requested separately.

What it measures	Oxygen dissolved in the water. This is different from BOD and COD (borewell report), which measure how much oxygen the pollutants in the water use up.
Method	IS 3025 (Part 38), Winkler azide modification, or a calibrated DO meter on site
Sampling	Fill a 300 ml glass BOD bottle completely, with no air bubbles. Fix the oxygen on site with $MnSO_4$ and alkaline iodide-azide. Analyse within 8 hours. A courier sample is not acceptable, so ask the lab to fix or measure it at the time of collection.
Interpretation	IS 10500 sets no limit. Good water is typically at least 5–6 mg/L. A low reading in tank or tap water means stagnant storage. A low reading together with high BOD (as in the borewell) confirms organic pollution.
Indicative cost	MPCB ₹250 per sample. Ask Varni Analytical for their add-on rate.
Also add	**Residual free chlorine** at the kitchen tap (should be at least 0.2 mg/L). It is the best routine indicator that the supply is protected against bacteria. It is cheap and can be checked on site.

5. Recommended actions

#	Action	Priority
1	Kitchen water (Oct 2026) is fit to drink. Continue normal use.	**Done**
2	Find out what changed between September and October, and keep borewell water **permanently separated** from drinking-water tanks and lines.	**Immediate**
3	Do not use borewell water for drinking or cooking. Inspect the borewell casing, cap and seal, and nearby drains and septic lines.	**Immediate**
4	Treat borewell water before any use: sediment or pressure sand filter, then activated carbon, then chlorination.	**Within 1 month**
5	Retest the kitchen water (bacteria) in about 4 weeks to confirm October's result. **Add Dissolved Oxygen and residual chlorine.** The lab should collect the sample.	**End Nov 2026**
6	Retest borewell water, with the full Dec-25 parameter list plus DO, after treatment or repairs.	**Within 1–2 months**
7	If the building has old GI pipes: test iron (and lead) once, because the soft, corrosive water can pick up metals from them.	Low
8	Routine programme: bacteriological test monthly for 3 months and then quarterly, full IS 10500 panel every 6 months, tank cleaning every 6 months.	Ongoing

Based on test reports issued by Varni Analytical LLP (VAL/2512/160070, VAL/2609/230075 and VAL/2610/270216). Results apply only to the samples tested. This summary is a technical interpretation and does not replace the original reports.